Stochastic Calculus and Stochastic Control

Gongqiu Zhang

SSE, CUHK-Shenzhen

Outline

1 Basics of Stochastic Calculus

2 Stochastic Control and Optimal Stopping

Random Variables

A random variable (r.v.) is a mapping from a sample space (Ω) to the real line ($\mathbb{R}$).

$$X : \Omega \to \mathbb{R}.$$

It is

- random, before the experiment;
- deterministic after the experiment is done: $X(\omega)$.

Example (Discrete): Let X be 1 if the outcome of rolling a die is odd, and 0 if the outcome is even. Then $X = 1_{\{\omega \in \{1,3,5\}\}}$.

Example (Continuous): Let Y be the outcome of spinning a wheel where Y is uniformly distributed on $[0, 1]$. Then $Y = \omega$ for $\omega \in [0, 1]$.

Stochastic Processes

Stochastic process is a collection of random variables indexed by time.

- Discrete time: $X_t, t \in \mathbb{Z}$ or $\mathbb{N}$.
- Continuous time: $X_t, t \in \mathbb{R}$ or $[0, T]$.

A stochastic process X_t

- is a random variable for a fixed t;
- is a deterministic function of t for a fixed ω: a trajectory/path.

Example of Stochastic Process

A random walk is a stochastic process X_t defined as

$$X_t = X_{t-1} + \epsilon_t,$$

where ϵ_t is a random variable independent of X_{t-1} and has the same distribution.

A Poisson process is a stochastic process N_t defined as

$$N_t = \sum_{i=1}^{\infty} 1_{\{\tau_i \leq t\}},$$

where τ_i is the i-th arrival time, a summation of i.i.d. exponential random variables with rate λ.

Limit Order Book as a Compound Poisson Process

A limit order book (LOB) aggregates all outstanding limit buy and sell orders for an asset. The LOB evolves as new orders arrive and existing orders are cancelled or executed.

A simple model for the LOB is the compound Poisson process:

- **Order arrivals:** Buy/sell orders arrive according to a Poisson process with intensity λ.
- **Order sizes:** Each order is associated with a random size J_i (i.i.d. non-negative random variables).

Let N_t be the number of orders that have arrived up to time t. Then the total volume added up to time t is

$$X_t = \sum_{i=1}^{N_t} J_i,$$

where $N_t \sim \text{Poisson}(\lambda t)$.

LOB as Compound Poisson Process (continued)

Now, we use stochastic processes to describe the dynamics at different price levels. Let p_k denote the k-th price level in the order book.

- $L_t^b(p_k)$ and $L_t^s(p_k)$: total outstanding buy and sell limit orders at price p_k at time t.
- **Limit Buy Order Arrivals:** At p_k, buy limit orders arrive via a (possibly compound) Poisson process with intensity λ_k^b. The total at p_k increases by random amounts $J_i^{b,(k)}$:

$$L_t^b(p_k) = L_0^b(p_k) + \sum_{i=1}^{N_t^b(p_k)} J_i^{b,(k)} - \text{depletions/cancellations}$$

where $N_t^b(p_k)$ has rate λ_k^b.

LOB as Compound Poisson Process (continued)

- **Limit Sell Order Arrivals:** Sell LOs at p_k arrive via a Poisson process with intensity λ_k^s and random sizes $J_i^{s,(k)}$.
- **Cancellations:** Limit orders at each price can be cancelled according to independent Poisson processes (reducing $L_t^b(p_k)$ or $L_t^s(p_k)$).
- **Market Order Arrivals:** Market buy and market sell orders arrive via independent Poisson processes with intensities μ^b (market buy) and μ^s (market sell).
  - A market buy order depletes the lowest available sell LOs, moving up price levels as needed.
  - A market sell order depletes outstanding buy LOs, starting at the highest bid.

LOB as Compound Poisson Process (continued)

Thus, for each price p_k, the dynamics of LOs (buy and sell) are:

$$L_t^b(p_k) = L_0^b(p_k) + \sum_{i=1}^{N_t^b(p_k)} J_i^{b,(k)} - D_t^b(p_k) - C_t^b(p_k),$$

$$L_t^s(p_k) = L_0^s(p_k) + \sum_{i=1}^{N_t^s(p_k)} J_i^{s,(k)} - D_t^s(p_k) - C_t^s(p_k),$$

where $D_t^b(p_k)$, $D_t^s(p_k)$ are cumulative depletions (market orders) and $C_t^b(p_k)$, $C_t^s(p_k)$ are cumulative cancellations up to time t at p_k.

Information Flows

The information up to time t means the events that have occurred up to time t which can be observed.

$$\mathcal{F}_t = \sigma(X_s : s \leq t).$$

where $\sigma(X_s : s \leq t)$ is the σ-algebra generated by the observable random variables X_s for $s \leq t$.

It is clear that

$$\mathcal{F}_s \subseteq \mathcal{F}_t, \qquad \text{for all } s \leq t.$$

because the information up to time s is a subset of the information up to time t for $s \leq t$.

A stochastic process X_t is called adapted to the filtration $\mathcal{F}_t$ if for all t, X_t is $\mathcal{F}_t$-measurable.

Brownian Motions

A stochastic process W_t is called a standard Brownian motion if it satisfies the following properties:

- $W_0 = 0$.
- Continuous paths: W_t is continuous for all $t \geq 0$ almost surely.
- Independent increments: for any $0 \leq t_0 < t_1 < \cdots < t_n$, the increments $W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent.
- Stationary increments: for any $0 \leq s < t$, $W_t - W_s$ has the same distribution as $W_{t-s} \sim \mathcal{N}(0, t-s)$.

A drifted Brownian motion is a Brownian motion with drift μ and volatility σ:

$$X_t = X_0 + \mu t + \sigma W_t.$$

The standard Brownian motion is a drifted Brownian motion with drift $\mu = 0$ and $\sigma = 1$.

Price Dynamics as a Brownian Motion

In a short time interval, the price change can be conveniently approximated by a Brownian motion.

$$dS_t = \mu dt + \sigma dW_t.$$

Challenge: Why is this a good approximation considering that the price process is always positive?

Martingales

A stochastic process X_t is called a martingale with respect to the filtration $\mathcal{F}_t$ if

$$\mathbb{E}[X_t|\mathcal{F}_s] = X_s, \qquad \text{for all } s \leq t.$$

Similar definitions can be given for submartingales and supermartingales.

A stochastic process X_t is called a submartingale (supermartingale) if

$$\mathbb{E}[X_t|\mathcal{F}_s] \geq (\leq) X_s, \qquad \text{for all } s \leq t.$$

respectively.

Examples of Martingales

Example: The Brownian motion W_t is a martingale.

Example: The conditional expectation of a random variable X given the information up to time t is a martingale.

$$\mathbb{E}[X|\mathcal{F}_t] = X_t.$$

Example: The running maximum of a random walk $M_t = \max_{0\leq s\leq t} X_s$ is a submartingale.

Example: $Z_t = X_t^2 - t$ is a martingale if X_t is a Brownian motion.

Markov Processes

A stochastic process X_t is called a Markov process with respect to the filtration $\mathcal{F}_t$ if for any function f, there exists a function g such that

$$\mathbb{E}[f(t, X_t)|\mathcal{F}_s] = g(s, X_s), \qquad \text{for all } s \leq t.$$

Example: A random walk X_t is a Markov process with respect to the filtration generated by itself.

Example: The Brownian motion W_t is a Markov process with respect to the filtration generated by itself.

Example: The running maximum of a random walk $M_t = \max_{0 \leq s \leq t} X_s$ is not a Markov process with respect to the filtration generated by X_t.

Ito Integrals

The Ito integral defines the integral of an adapted stochastic process (random function) with respect to a Brownian motion.

$$\int_0^t \theta_s dW_s = \lim_{|\Pi|\to 0} \sum_{i=1}^{n} \theta_{t_{i-1}}(W_{t_i} - W_{t_{i-1}}).$$

where Π is a partition of $[0, t]$: $0 = t_0 < t_1 < \cdots < t_n = t$.

The quadratic variation of the Ito integral is

$$\left[\int_0^{\cdot} \theta_s dW_s, \int_0^{\cdot} \theta_s dW_s\right]_t = \lim_{|\Pi|\to 0} \sum_{i=1}^{n} \theta_{t_{i-1}}^2 (t_i - t_{i-1}) = \int_0^t \theta_s^2 ds.$$

PnL as a Ito Integral

The PnL of a trading strategy can be expressed as a Ito integral.

$$dPnL_t = \theta_t dW_t,$$

if dW_t models the price change.

The adaptedness of θ_t ensures that the trader cannot use future information to make decision. In this sense, the PnL is a martingale.

The QV of Brownian Motion in an Interval

The quadratic variation of the Brownian motion in an interval $[s, t]$ is given by

$$[W, W]_{s,t} = \int_s^t 1 ds = t - s.$$

Proof: We first partition the interval $[s, t]$ into n subintervals: $s = t_0 < t_1 < \cdots < t_n = t$. Then we have

$$[W, W]_{s,t} = \lim_{|\Pi| \to 0} \sum_{i=1}^{n} (W_{t_i} - W_{t_{i-1}})^2.$$

Since the increments of the Brownian motion are independent, we have

$$\mathbb{E}\left[(W_{t_i} - W_{t_{i-1}})^2\right] = t_i - t_{i-1}.$$

Moreover, for the variance of summed squared increments, we have

$$\begin{aligned} \text{Var}\left[\sum_{i=1}^{n} (W_{t_i} - W_{t_{i-1}})^2\right] &= \sum_{i=1}^{n} \text{Var}\left[(W_{t_i} - W_{t_{i-1}})^2\right] \\ &= \sum_{i=1}^{n} 2(t_i - t_{i-1})^2 \leq 2|\Pi|(t - s) \to 0. \end{aligned}$$

Properties of Ito Integrals

The Ito integral has the following properties:

- Linearity:

$$\int_0^t (\alpha\theta_s + \beta\psi_s)dW_s = \alpha\int_0^t \theta_s dW_s + \beta\int_0^t \psi_s dW_s.$$

- Martingale property: $\int_0^t \theta_s dW_s$ is a martingale with respect to the filtration $\mathcal{F}_t$.
- Quadratic variation: $[\int_0^\cdot \theta_s dW_s, \int_0^\cdot \theta_s dW_s]_t = \int_0^t \theta_s^2 ds$.
- Ito isometry: $\mathbb{E}\left[\left(\int_0^t \theta_s dW_s\right)^2\right] = \mathbb{E}\left[\int_0^t \theta_s^2 ds\right]$.

Ito Processes

An Ito process is a stochastic process of the form

$$X_t = X_0 + \int_0^t \theta_s dW_s + \int_0^t \psi_s ds,$$

where θ_s and ψ_s are adapted processes.

The quadratic variation of the Ito process is

$$[X, X]_t = \int_0^t \theta_s^2 ds.$$

In differential form, it is

$$dX_t dX_t = \theta_t^2 dt,$$

where we use the convention that

$$dW_t dW_t = dt, \quad dtdt = 0, \quad \text{and} \quad dW_t dt = 0.$$

Example: Permanent Price Impact

Suppose the price impact is permanent:

$$dS_t = (\mu + g(\nu_t))dt + \sigma dW_t.$$

where $g(\nu_t)$ is the permanent price impact function and ν_t is the trading rate (an adapted process).

S_t is an Ito process:

$$S_t = S_0 + \int_0^t (\mu + g(\nu_s))ds + \int_0^t \sigma dW_s.$$

Ito Formula

The Ito formula is a generalization of the chain rule for deterministic functions to stochastic functions.

$$f(t, X_t) = f(t, X_0) + \int_0^t f_t(s, X_s)ds + \int_0^t f_x(s, X_s)dX_s + \frac{1}{2}\int_0^t f_{xx}(s, X_s)[X, X]_s ds.$$

In differential form, it is

$$df(t, X_t) = f_t(t, X_t)dt + f_x(t, X_t)dX_t + \frac{1}{2}f_{xx}(t, X_t)d[X, X]_t.$$

Example: Exponential Martingale $e^{-\sigma W_t - \frac{1}{2}\sigma^2 t}$

Stochastic Differential Equations

A stochastic differential equation takes the following form:

$$dX_t = \theta(t, X_t)dt + \psi(t, X_t)dW_t,$$

where θ and ψ are specific functions.

It is also known as a one-dimensional diffusion process, which has the Markov property.

Example: Geometric Brownian Motion

Suppose X_t is a geometric Brownian motion:

$$dX_t = \mu X_t dt + \sigma X_t dW_t.$$

Then the Ito formula gives

$$df(t, X_t) = f_t(t, X_t)dt + f_x(t, X_t)dX_t + \frac{1}{2} f_{xx}(t, X_t)d[X, X]_t.$$

where $f(t, X_t) = \ln X_t$.

Therefore,

$$d \ln X_t = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t.$$

The solution to the geometric Brownian motion is

$$X_t = X_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t\right).$$

Example: OU Process $dX_t = \kappa(\theta - X_t)dt + \sigma dW_t$

The Feynman-Kac Theorem

Under some conditions, the conditional expectation of a function of a stochastic process:

$$\mathbb{E}_{t,x}[e^{-\int_t^T r(s,X_s)ds} g(X_T)] = f(t,x),$$

satisfies the following partial differential equation:

$$\begin{cases} f_t(t,x) + \theta(t,x) f_x(t,x) + \frac{1}{2}\psi^2(t,x) f_{xx}(t,x) - r(t,x) f(t,x) = 0, \\ f(T,x) = g(x). \end{cases}$$

Example: Calculating $\mathbb{E}[W_T^4]$

The Infinitesimal Generator of Diffusion Processes

The infinitesimal generator of a Markov process is defined as

$$\mathcal{L}f(t,x) = \lim_{s \downarrow t} \frac{\mathbb{E}_{t,x}[f(t,X_s) - f(t,x)]}{s-t}.$$

For a diffusion process, the infinitesimal generator is given by

$$\mathcal{L}f(t,x) = \theta(t,x)f_x(t,x) + \frac{1}{2}\psi^2(t,x)f_{xx}(t,x).$$

Given this notion, the PDE for the conditional expectation can be written as

$$f_t(t,x) + \mathcal{L}f(t,x) - r(t,x)f(t,x) = 0.$$

Multidimensional Brownian Motions

A multidimensional Brownian motion is a vector of Brownian motions:

$$W_t = (W_{1,t}, W_{2,t}, \ldots, W_{d,t}).$$

When applying the Ito formula to a multidimensional Brownian motion, we use the convention that

$$dW_{i,t} dW_{j,t} = \delta_{ij} dt,$$

where δ_{ij} is the Kronecker delta.

Multivariate Ito Processes

A m-dimensional multivariate Ito process is a vector of Ito processes:

$$X_t = X_0 + \int_0^t \theta_s dW_s + \int_0^t \psi_s ds,$$

where $\theta_s \in \mathbb{R}^{m \times d}$ and $\psi_s \in \mathbb{R}^m$.

When applying the Ito formula to a multivariate Ito process, we use the convention that

$$dW_{i,t} dW_{j,t} = \delta_{ij} dt,$$

where δ_{ij} is the Kronecker delta.

Multivariate Ito Formula

The Ito formula for a multivariate Ito process is given by

$$f(t, X_t) = f(0, X_0) + \int_0^t f_t(s, X_s)ds + \int_0^t f_x(s, X_s)^\top dX_s + \frac{1}{2}\int_0^t dX_s^\top f_{xx}(s, X_s)dX_s.$$

In differential form, it is

$$df(t, X_t) = f_t(t, X_t)dt + f_x(t, X_t)^\top dX_t + \frac{1}{2}dX_t^\top f_{xx}(t, X_t)dX_t.$$

Multivariate Stochastic Differential Equations

A m-dimensional multivariate stochastic differential equation takes the following form:

$$dX_t = \theta(t, X_t)dt + \psi(t, X_t)dW_t,$$

where $\theta : \mathbb{R} \times \mathbb{R}^m \to \mathbb{R}^m$ and $\psi : \mathbb{R} \times \mathbb{R}^m \to \mathbb{R}^{m \times d}$ are specific functions.

It has the Markov property.

The Multivariate Feynman-Kac Theorem

Under some conditions, the conditional expectation of a function of a multivariate Ito process:

$$\mathbb{E}_{t,x}[g(X_T)] = f(t,x),$$

satisfies the following partial differential equation:

$$\begin{cases} f_t(t,x) + \mathcal{L}f(t,x) - r(t,x)f(t,x) = 0, \\ f(T,x) = g(x). \end{cases}$$

Here

$$\mathcal{L}f(t,x) = \theta(t,x)^\top f_x(t,x) + \frac{1}{2}\text{trace}(\psi(t,x)^\top f_{xx}(t,x)\psi(t,x)).$$

Example: Self-Financing Portfolio

A self-financing portfolio is a portfolio that is financed by the initial wealth and the returns from the portfolio itself, i.e.,

$$dX_t = r(X_t - \pi_t S_t)dt + \pi_t dS_t,$$

where π_t is the amount of the asset held in the portfolio at time t (an adapted process), and S_t is the price of the asset at time t. r is the risk-free rate.

Suppose that the asset price follows an SDE:

$$dS_t = \mu(t, S_t)dt + \sigma(t, S_t)dW_t.$$

Then the portfolio value process is given by

$$dX_t = r(X_t - \pi_t S_t)dt + \pi_t(\mu(t, S_t)S_t dt + \sigma(t, S_t)S_t dW_t),$$

Question: what is the infinitesimal generator of (X_t, S_t) conditioned on π_t?

Poisson Processes

A Poisson process is a stochastic process that counts the number of events that occur in a fixed interval of time.

$$N_t = \sum_{i=1}^{\infty} 1_{\{\tau_i \le t\}},$$

where τ_i is the arrival time of the i-th event, a summation of i.i.d. exponential random variables with rate λ.

The probability of having k events in a time interval with duration τ is given by ($\tau = t - s$)

$$\mathbb{P}[N_t - N_s = k] = e^{-\lambda\tau} \frac{(\lambda\tau)^k}{k!},$$

where $\tau = t - s$.

The increments are independent, and stationary (i.e., $N_t - N_s$ has the same distribution as N_{t-s} for all $t \ge s$).

Compound Poisson Processes

A compound Poisson process is a stochastic process in the form:

$$C_t = \sum_{i=1}^{N_t} Y_i,$$

where Y_i is a sequence of i.i.d. random variables, and N_t is a Poisson process with rate λ.

The Infinitesimal Generator of Compound Poisson Processes

The infinitesimal generator of a compound Poisson process is given by

$$\mathcal{L}f(t,x) = \int_{\mathbb{R}} (f(t,x+y) - f(t,x))\nu(dy),$$

where $\nu(dy)/\lambda$ is the distribution of the jump size.

Stochastic Differential Equations with Jump Processes

A stochastic differential equation with a jump process takes the following form:

$$dX_t = \theta(t, X_t)dt + \psi(t, X_t)dW_t + \beta(t, X_t)dC_t,$$

where C_t is a compound Poisson process.

The infinitesimal generator is given by:

$$\mathcal{L}f(t,x) = \theta(t,x)f_x(t,x) + \frac{1}{2}\psi^2(t,x)f_{xx}(t,x) + \int_{\mathbb{R}} (f(t, x+\beta(t,x)y) - f(t,x))\nu(dy).$$

Stochastic Jump Intensity

It is often beneficial to model the jump intensity as a stochastic process, for example:

$$d\lambda_t = \kappa(\theta - \lambda_t)dt + \sigma\sqrt{\lambda_t}dB_t,$$

such that $dB_t dW_t = \rho dt$.
In this case, (X, λ) jointly forms a Markov process with infinitesimal generator:

$$\begin{aligned}\mathcal{L}f(t,x,\lambda) = \theta(t,x)f_x(t,x) + \frac{1}{2}\psi^2(t,x)f_{xx}(t,x) \\ + \lambda\int_{\mathbb{R}}(f(t,x+\beta(t,x)y) - f(t,x))\overline{\nu}(dy) \\ + \kappa(\theta-\lambda)f_\lambda(t,x,\lambda) + \frac{1}{2}\sigma^2\lambda f_{\lambda\lambda}(t,x,\lambda) \\ + \rho\sigma\psi(t,x)\sqrt{\lambda}f_{\lambda,x}(t,x,\lambda).\end{aligned}$$

Outline

1 Basics of Stochastic Calculus

2 Stochastic Control and Optimal Stopping

Stochastic Control Problems

A stochastic control problem is a problem of finding an optimal control that maximizes the expected reward.

$$\max_{u \in \mathcal{U}_{0,T}} \mathbb{E}\left[\int_0^T F(t, X_t^u, u_t)dt + G(T, X_T^u)\right],$$

where X_t is a Markov process, and u_t is a control process.

The solution to the stochastic control problem is the optimal control policy.

The Dynamic Programming Principle

The dynamic programming principle (DPP) states that the optimal control policy is still optimal from any time t onwards.

Let

$$H(t,x) = \sup_{u \in \mathcal{U}_{t,T}} H^u(t,x) = \mathbb{E}_{t,x}\left[\int_t^T F(s, X_s^u, u_s)ds + G(T, X_T^u)\right],$$

Then the DPP yiels that for any (t,x) and $\tau \in [t,T]$,

$$H(t,x) = \sup_{u \in \mathcal{U}_{t,T}} \mathbb{E}_{t,x}\left[\int_t^\tau F(s, X_s^u, u_s)ds + H(\tau, X_\tau^u)\right].$$

The Hamilton-Jacobi-Bellman (HJB) Equation

Moving the right-hand side to the left-hand side and sending $\tau \downarrow t$, we get the Hamilton-Jacobi-Bellman (HJB) equation:

$$\sup_u \{H_t(t,x) + \mathcal{L}^u H(t,x) + F(t,x,u)\} = 0.$$

The terminal condition:

$$H(T,x) = G(T,x).$$

The Case with Discounting

Suppose the discount rate is $r(t,x)$:

$$\max_{u\in\mathcal{U}_{0,T}} \mathbb{E}\left[\int_0^T e^{-\int_0^s r(v,X_v^u)dv} F(s,X_s^u,u_s)ds + e^{-\int_0^T r(v,X_v^u)dv} G(T,X_T^u)\right].$$

Correspondingly, Let

$$H(t,x) = \mathbb{E}_{t,x}\left[\int_t^T e^{-\int_t^s r(v,X_v^u)dv} F(s,X_s^u,u_s)ds + e^{-\int_t^T r(v,X_v^u)dv} G(T,X_T^u)\right]$$

The HJB equation with discounting is given by:

$$\sup_u\{H_t(t,x) + \mathcal{L}^u H(t,x) + F(t,x,u) - r(t,x)H(t,x)\} = 0.$$

The terminal condition:

$$H(T,x) = G(T,x).$$

Example: Optimal Portfolio Selection

We consider the Merton's problem:

$$\max_{\pi \in \mathcal{A}} \mathbb{E}\left[U(X_T^\pi)\right],$$

where

- X_t^π is the portfolio value process;
- π_t is the amount of money invested in the asset at time t (an adapted process);
- $U(x) = -e^{-\gamma x}$ $(\gamma > 0)$ is the utility function.

Example: Optimal Portfolio Selection (continued)

Suppose the asset price follows a geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ is the drift rate and σ is the volatility.

Then the portfolio value process is given by

$$\begin{aligned} dX_t^\pi &= r(X_t^\pi - \pi_t)dt + \frac{\pi_t}{S_t}(\mu S_t dt + \sigma S_t dW_t) \\ &= (rX_t^\pi + \pi_t(\mu - r))dt + \pi_t \sigma dW_t. \end{aligned}$$

The infinitesimal generator of X_t^π conditioned on π_t is given by

$$\mathcal{L}^\pi f(t,x) = (rx + \pi(\mu - r))f_x(t,x) + \frac{1}{2}\pi^2\sigma^2 f_{xx}(t,x).$$

Example: Optimal Portfolio Selection (continued)

Let $H(t,x) = \sup_{\pi \in \mathcal{A}_{[t,T]}} \mathbb{E}_{t,x}\left[U(X_T^\pi)\right]$.

The HJB equation for the optimal portfolio selection problem is given by

$$\sup_\pi \{H_t(t,x) + (rx + \pi(\mu - r))H_x(t,x) + \frac{1}{2}\pi^2\sigma^2 H_{xx}(t,x)\} = 0.$$

The terminal condition:

$$H(T,x) = -e^{-\gamma x}.$$

Let $\lambda = (\mu - r)/\sigma$. We have that

$$\pi^* = -\frac{\lambda}{\sigma}\frac{(H_x(t,x))^2}{H_{xx}(t,x)}.$$

Example: Optimal Portfolio Selection (continued)

The HJB equation reduces to

$$H_t(t,x) - \frac{\lambda}{2\sigma}\frac{(H_x(t,x))^2}{H_{xx}(t,x)} + rxH_x(t,x) = 0.$$

Suppose that $H(t,x) = \alpha(t)e^{-\gamma\beta(t)x}$. Then

$$\alpha'(t) = \lambda\alpha(t)/(2\sigma),\ \alpha(T) = 1,$$
$$\beta'(t) = -r\beta(t), \beta(T) = \gamma.$$

The solutions are given by

$$\alpha(t) = e^{-\lambda(T-t)/(2\sigma)},\ \beta(t) = e^{r(T-t)}.$$

The optimal portfolio is given by

$$\pi^* = \frac{\lambda}{\gamma\sigma}e^{-r(T-t)}.$$

Example: Maximizing Expected Wealth using Round-Trip Trades

An agent uses a market order to purchase an asset at time t and then submit a limit sell order at depth u_t. Suppose the midprice is S_t, the profit of each successful round-trip trade is

$$(S_t + u_t) - (S_t + \Delta/2) = u_t - \Delta/2,$$

where Δ is the bid-ask spread assumed to be constant.

The controlled wealth process X_t^u satisfies that:

$$dX_t^u = (u_t - \Delta/2)dN_t^u,$$

where N_t^u is a counting process with intensity $\lambda_t^u = \Lambda e^{-\kappa u_t}$.

The objective is to maximize the expected wealth: $\max_{u\in\mathcal{U}} \mathbb{E}\left[X_T^u\right]$.

Example: Maximizing Expected Wealth using Round-Trip Trades (continued)

The HJB equation for the optimal stopping problem is given by:

$$\max_u \{H_t(t,x) + \Lambda e^{-\kappa u}(H(t,x+u-\Delta/2) - H(t,x))\} = 0.$$

Suppose that $H(t,x) = x + h(t)$. Then we get that

$$h'(t) + \Lambda \sup_{u \geq 0} e^{-\kappa u}(u - \Delta/2) = 0.$$

The FO implies that

$$u^* = \frac{1}{\kappa} + \frac{\Delta}{2}, \; h'(t) = -\frac{\Lambda}{\kappa} e^{-(1+\kappa\Delta/2)} \Rightarrow h(t) = \frac{\Lambda}{\kappa} e^{-(1+\kappa\Delta/2)}(T-t).$$

Optimal Stopping

The optimal stopping problem is a problem of finding the optimal stopping time to maximize the expected reward.

$$H(t,x) = \max_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}\left[\int_t^\tau F(s, X_s)ds + G(\tau, X_\tau)\right],$$

where $\mathcal{T}_{t,T}$ is the set of all stopping times in $[t, T] \cup \{\infty\}$ (the agent can only base the decision on the information available at time t).

The Dynamic Programming Principle for Optimal Stopping

The DPP for optimal stopping: for any $t' \in [t, T]$

$$H(t,x) = \max_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}_{t,x}\left[\int_t^{t'} F(s, X_s)ds + H(t', X_{t'}^u)1_{\{\tau \geq t'\}} + G(\tau, X_\tau)1_{\{\tau < t'\}}\right]$$

The above equation implies that

$$\max\{H_t(t,x) + \mathcal{L}H(t,x) + F(t,x), G(t,x) - H(t,x)\} = 0.$$

The terminal condition:

$$H(T,x) = \max(0, G(T,x)).$$

The Case with Discounting for Optimal Stopping

Suppose the discount rate is $r(t,x)$:

$$H(t,x) = \max_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}_{t,x}\left[\int_t^\tau e^{-\int_t^s r(v,X_v)dv} F(s,X_s)ds + e^{-\int_t^\tau r(v,X_v)dv} G(\tau,X_\tau)\right].$$

The HJB equation with discounting is given by:

$$\max\{H_t(t,x) + \mathcal{L}H(t,x) + F(t,x) - r(t,x)H(t,x), G(t,x) - H(t,x)\} = 0.$$

The terminal condition:

$$H(T,x) = G(T,x).$$

Example: Optimal Stopping

Consider the problem of finding the optimal stopping time to maximize the expected reward ($T = \infty$):

$$H(t,x) = \max_{\tau \in \mathcal{T}_t} \mathbb{E}\left[e^{-r\tau} \max(0, K - X_\tau)\right],$$

where X is a geometric Brownian motion:

$$dX_t = rX_t dt + \sigma X_t dW_t.$$

The HJB equation is given by:

$$\max\{rxH_x + \frac{1}{2}\sigma^2 x^2 H_{xx} - rH, K - x - H\} = 0.$$

Example: Optimal Stopping (continued)

The solution to the HJB equation is given by:

$$H(x) = \begin{cases} K - x, & x \leq L, \\ ax^b, & x > L. \end{cases}$$

Then $rb + \frac{1}{2}\sigma^2 b(b-1) = r$, we take the negative root: $b = -2r/\sigma^2$.

a and L can be determined by the smooth pasting conditions:
$H_x(L-) = H_x(L+)$ and $H(L-) = H(L+)$.

Example: Selling at the Peak

Concept

The Russian Option (Shepp & Shiryaev, 1993) addresses the classic investor dilemma: *"I want to sell at the peak."*

Since we cannot foresee the future peak, we maximize the expected discounted value of the historical maximum price up to the stopping time.

$$\max_{\tau} \mathbb{E}\left[e^{-\lambda\tau} M_\tau\right].$$

Example: Selling at the Peak (continued)

Let the asset price S_t follow a geometric Brownian motion:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad S_0 = s$$

Define the Running Maximum M_t:

$$M_t = \max\left\{m, \sup_{0\le u\le t} S_u\right\}, \quad m \ge s$$

We seek the value function $V(s,m)$:

$$V(s,m) = \sup_{\tau} \mathbb{E}\left[e^{-\lambda\tau} M_\tau \mid S_0 = s, M_0 = m\right].$$

We assume $\lambda > \mu$ to ensure the value is finite.

Dimensionality Reduction

The problem has two state variables (s, m). Note the scaling property:

$$V(k \cdot s, k \cdot m) = k \cdot V(s, m)$$

The function is homogeneous of degree 1.

Define the drawdown ratio x:

$$x = s/m, \quad x \in (0, 1]$$

We reduce the PDE to an ODE by postulating:

$$V(s, m) = m \cdot W(x)$$

where $W(x)$ is the normalized value function.

Deriving the ODE for $W(x)$ (Step 1)

Inside the continuation region (where we hold the asset), $S_t < M_t$, so $dM_t = 0$. The Hamilton-Jacobi-Bellman (HJB) equation is:

$$\mathcal{L}V - \lambda V = \frac{1}{2}\sigma^2 s^2 \frac{\partial^2 V}{\partial s^2} + \mu s \frac{\partial V}{\partial s} - \lambda V = 0.$$

Substitute $V = mW(s/m)$:

$$\frac{\partial V}{\partial s} = mW'(x) \cdot \frac{1}{m} = W'(x)$$
$$\frac{\partial^2 V}{\partial s^2} = W''(x) \cdot \frac{1}{m}$$

Deriving the ODE for $W(x)$ (Step 2)

Plugging derivatives back into the HJB:

$$\frac{1}{2}\sigma^2 s^2 \left(\frac{W''(x)}{m}\right) + \mu s W'(x) - \lambda m W(x) = 0.$$

Divide by m and use $s = mx$:

$$\frac{1}{2}\sigma^2 (mx)^2 \frac{W''(x)}{m^2} + \mu (mx)\frac{W'(x)}{m} - \lambda W(x) = 0.$$

The ODE for $W(x)$

$$\frac{1}{2}\sigma^2 x^2 W''(x) + \mu x W'(x) - \lambda W(x) = 0.$$

Solving the ODE for $W(x)$

We guess a solution of the form $W(x) = x^p$.

Characteristic Equation:

$$\frac{1}{2}\sigma^2 p(p-1) + \mu p - \lambda = 0$$

$$\sigma^2 p^2 + (2\mu - \sigma^2)p - 2\lambda = 0$$

Roots $p_1 > p_2$:

$$p_{1,2} = \frac{-(2\mu - \sigma^2) \pm \sqrt{(2\mu - \sigma^2)^2 + 8\sigma^2\lambda}}{2\sigma^2}$$

Since $\lambda > 0$, we have $p_1 > 1$ and $p_2 < 0$.

$$W(x) = C_1 x^{p_1} + C_2 x^{p_2}$$

Applying the Boundary Conditions

We need to find the optimal stopping boundary $\psi \in (0,1)$. We continue to hold while $x > \psi$.

1. **Value Matching (at $x = \psi$):** When we stop, payoff is M. So $V = m \implies mW(\psi) = m$. So,
$$W(\psi) = 1.$$
2. **Smooth Pasting (at $x = \psi$):** Optimality requires smooth transition. The payoff M is constant w.r.t S. So,
$$W'(\psi) = 0.$$
3. **Reflection at the Maximum (at $x = 1$):** At the maximum ($S_t = M_t$), M_t increases. The constraint prevents x exceeding 1.
$$\left.\frac{\partial V}{\partial m}\right|_{s=m} = 0 \implies W(1) - 1 \cdot W'(1) = 0.$$

Solving the System of Equations

We have 3 equations for unknowns C_1, C_2, ψ:

1. $C_1\psi^{p_1} + C_2\psi^{p_2} = 1.$
2. $C_1 p_1 \psi^{p_1-1} + C_2 p_2 \psi^{p_2-1} = 0.$
3. $C_1(1-p_1) + C_2(1-p_2) = 0$ (from Reflection at $x = 1$).

From (3), express C_2:

$$C_2 = C_1 \frac{p_1 - 1}{1 - p_2}.$$

Substitute into (2) (multiplied by ψ):

$$C_1 p_1 \psi^{p_1} + \left(C_1 \frac{p_1 - 1}{1 - p_2} \right) p_2 \psi^{p_2} = 0.$$

The Optimal Threshold

Rearranging the previous equation yields the closed-form solution for ψ^*:

$$\frac{\psi^{p_1}}{\psi^{p_2}} = \frac{-p_2(p_1-1)}{p_1(1-p_2)}.$$

The Analytic Solution

The optimal strategy is to stop when $S_t \leq \psi^* M_t$, where:

$$\psi^* = \left(\frac{p_2(1-p_1)}{p_1(1-p_2)}\right)^{\frac{1}{p_1-p_2}}.$$

This defines a geometric trailing stop.

Numerical Test: SPY Data

Let us calibrate to current market estimates:

- Drift (μ): 8.5% (0.085).
- Volatility (σ): 16% (0.16).
- Discount (λ): 10% (0.10) *(Impatience factor)*.

Step 1: Characteristic Equation Roots:

$$0.0256p^2 + 0.1444p - 0.20 = 0.$$

$$p_1 \approx 1.150, \quad p_2 \approx -6.79.$$

Numerical Test: Results

Step 2: Calculate Threshold:

$$\psi^* = \left(\frac{-6.79(1-1.150)}{1.150(1-(-6.79))} \right)^{\frac{1}{1.150-(-6.79)}}.$$

$$\psi^* = \left(\frac{-6.79(-0.150)}{1.150(7.79)} \right)^{\frac{1}{7.94}} \approx \mathbf{0.76}.$$

Interpretation

The optimal strategy for these parameters is to sell the SPY portfolio if it drops **24%** below its all-time high achieved during the holding period.

Combining Stochastic Control and Optimal Stopping

The problem is to find the optimal control policy up to the stopping time that maximizes the expected reward.

$$H(t,x) = \max_{\tau \in \mathcal{T}_{t,T}} \sup_{u \in \mathcal{U}_{t,\tau}} \mathbb{E}_{t,x}\left[\int_t^\tau F(s, X_s^u, u_s)ds + G(\tau, X_\tau^u)\right].$$

The HJB equation for the combined problem is given by:

$$\max\{H_t(t,x) + \sup_u \mathcal{L}^u H(t,x) + F(t,x,u), G(t,x) - H(t,x)\} = 0.$$